Hybrid Reward Architecture for Reinforcement Learning

Summary

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1 Problem Setup

Traditional reinforcement learning relies on a single scalar reward function to guide learning, but many complex problems involve multiple distinct reward signals that need to be balanced. For example, in game playing, an agent might need to consider score, survival, exploration, and resource management simultaneously. This paper introduces the Hybrid Reward Architecture (HRA), which decomposes complex reward functions into multiple specialized components and combines them for decision-making.

2 Thesis

The central thesis is that decomposing complex reward functions into multiple specialized "reward heads" allows for more effective learning and better generalization than using a single monolithic reward function. The authors argue that different aspects of the reward signal can be learned more efficiently by specialized components that focus on specific sub-tasks.

3 Key Concepts

Reward Heads: Separate learning components that each focus on a specific aspect of the overall reward function, such as score maximization, ghost avoidance, or exploration.

General Value Functions (GVFs): Prediction functions that estimate future cumulative signals for different reward components, allowing the system to learn multiple value functions simultaneously.

Aggregator: A mechanism that combines the outputs from multiple reward heads to make final action decisions, typically through weighted summation.

Diversification Head: A specialized component that encourages exploration by providing random values for actions, preventing deterministic behavior patterns.

Targeted Exploration Head: A component that prioritizes exploration of less-visited state-action pairs using upper confidence bound-inspired mechanisms.

4 Methods

The authors develop the Hybrid Reward Architecture with several key components:

1. **Reward Decomposition:** Complex reward functions are decomposed into multiple specialized components, each handled by a separate reward head.

2. **General Value Functions:** Each reward head learns a GVF that predicts future cumulative returns for its specific reward signal.

3. **Aggregation Mechanism:** The outputs from all reward heads are combined, typically through normalized summation, to make final action decisions.
4. **Exploration Components:** Specialized heads for diversification (random exploration) and targeted exploration (prioritizing unvisited states) are added to ensure comprehensive exploration.
5. **Executive Memory:** An optional component that records successful action sequences for replay in similar situations, enabling level-specific learning.

The architecture allows different components to use different discount factors and learning rates, enabling fine-grained control over the learning process.

5 Results

The authors demonstrate HRA's effectiveness on several domains:

- **Ms. Pac-Man:** HRA achieves scores of 15,000-20,000 points, significantly outperforming A3C baselines and approaching human-level performance.
- **Component Analysis:** Individual reward heads successfully learn their specific tasks (score maximization, ghost avoidance, pellet collection) and combine effectively.
- **Exploration Benefits:** The combination of diversification and targeted exploration heads leads to more efficient exploration than traditional epsilon-greedy methods.
- **Level Passing:** With the executive memory component, HRA can successfully pass entire levels in Ms. Pac-Man, demonstrating long-term planning capabilities.
- **Robustness:** The architecture performs well across different hyperparameter settings, showing good generalization and stability.

The results show that reward decomposition enables more efficient learning and better performance than monolithic reward approaches.

6 Significance

This work introduces an important paradigm for handling complex reward structures in reinforcement learning:

- **Modular Design:** Provides a modular approach to reward function design that allows different aspects of complex problems to be learned separately.
- **Interpretability:** The decomposition makes the learning process more interpretable, as each component has a clear purpose and can be analyzed independently.
- **Scalability:** The architecture can scale to complex problems with many different reward signals that would be difficult to balance in a single reward function.
- **Exploration Innovation:** Introduces novel exploration mechanisms that are more sophisticated than traditional random exploration approaches.

- **Practical Impact:** Demonstrates significant performance improvements on challenging domains, showing the practical value of reward decomposition.

The paper established reward decomposition as a viable and effective approach for complex reinforcement learning problems, influencing subsequent work on multi-objective and hierarchical reinforcement learning.